

Werner Bisschoff

SOFTWARE ENGINEER

Cape Town, Western Cape, South Africa

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Summary

Software engineer with 5+ years of experience bridging hardware and full-stack development. I combine technical depth in C++ and Python with a rigorous, agentic engineering methodology. My focus is on resilient system design and productivity-enhancing AI workflows, governed by security, processes, and specification-driven development.

Experience

Divergent Tabletop

Cape Town

FOUNDER AND HOST

Jul 2025 – Present

- Founded and facilitate a neurodivergent-focused peer community, managing event operations and a WhatsApp-based community.
- Conflict resolution and management in a neurodiverse context.
- Applied systems thinking to solve community management challenges through software solutions.

FARO Africa

Cape Town

FULL-STACK SOFTWARE ENGINEER

Aug 2024 – Nov 2025

- Extended ERPNext using Python/JavaScript to improve workflows, pricing logic, and operational reporting (SQL)
→ **reduced manual reporting time and improved data accuracy**
- Built mobile application in Expo, including NFC (ISO 14443-4 APDUs) for e-paper price tags and card operations
→ **enabled real-time price updates with fewer tagging errors**
- Migrated internal mobile Retool workflows to Expo
→ **improved performance and enhanced long-term maintainability**
- Developed and maintained C# APIs supporting internal systems.
- Provisioned AWS infra with Pulumi and deployed services including Inngest and PayloadCMS.
- Diagnosed and resolved issues in a large existing ERPNext installation.
- Introduced LLM-assisted development workflows
→ **improved debugging speed and code review throughput, enabling faster iteration**

Ingenics Digital GmbH (through ViVa Outsourcing)

Remote Work

EMBEDDED SOFTWARE ENGINEER

Mar 2023 – May 2024

- Designed an event-driven finite state machine for an I2C-based embedded system using C++ and FreeRTOS
→ **created maintainable in-house architecture leading to fewer bugs and quicker development**
- Developed ESP32 applications using C/C++ and ESP-IDF
- Integrated a configurable low-energy BLE stack for device communication
- Built Python-based tooling for serial/BLE communication, including client-facing test executables
→ **accelerated testing and debugging workflows**
- Created Python hardware mocks for rapid iteration and early-stage testing
→ **enabled faster development cycles with fewer hardware dependencies**
- Managed fast, reliable data interchange using a compact TinyFrame binary protocol
- Implemented asynchronous communication workflows with Python and Pytest
- Developed an active object within the QP Real-Time Embedded Framework (with QSPY) to simulate device behaviour
- Integrated a configurable BLE stack for device communication and implemented FOTA firmware updates over BLE

UMAN Technologies

Century City, Cape Town

SOFTWARE DEVELOPER

Mar 2021 – Dec 2022

- Creating and maintaining Docker containers for development and CI/CD testing
→ **improved development environment consistency**
- Implementing and testing new services using RPC based on the SOME/IP protocol, as well as using *perf* to reduce performance bottlenecks
- Implementing IPC/RPC in existing C++ programs and Python scripts using Cap'n Proto and pycapnp
- Implementing a node tree to expose process-related variables and function calls to the IPC interface
- Analysing TCP/UDP traffic with Wireshark
- Leading a small team using AGILE development practices, including onboarding and mentoring new software developers

North-West University

Potchefstroom

JUNIOR LECTURER

Feb 2020 – Dec 2020

- Lecturing Python and C++ programming for *Introduction to Programming* for first year IT students in both remote and in-person settings

Education

North-West University

B.ENG. COMPUTER AND ELECTRONIC ENGINEERING

Potchefstroom

2020

- Focus on embedded systems, software engineering, and electronic design
- Developing an Android app with Kotlin to emulate an ISO 14443 protocol-based NFC payment system
- Developing microcontroller logic with C and the STM32 system as well as utilizing STM32CubeMX
- Implementing a PID controller with an Arduino to control a DC motor’s voltage and speed
- Cleaning and analysing data from large spreadsheets with Python and Pandas, utilizing linear regression, correlation and machine learning

Projects

Community Knowledge Base

DIVERGENT TABLETOP WIKI

Cape Town

Jun 2025 – Present

- Built a community wiki using Astro, Elixir, and Docker for knowledge management
- Documented event frameworks, onboarding processes, and communication best practices
- Created tooling for content management and community operations

Event-Driven FSM for Embedded Systems

INGENICS DIGITAL GMBH

Mar 2023 – May 2024

- Designed event-driven finite state machine for I2C-based embedded system using C++ and FreeRTOS
- Solved complex state management challenges in real-time embedded environment
- **Outcome:** Maintainable in-house architecture leading to fewer bugs and quicker development cycles

Skills

Systems Architecture	Event-driven FSM RPC/IPC (Cap’n Proto, SOME/IP) State machines Node trees
Testing & Quality	Unit testing (Pytest) Hardware mocks Test executables Code coverage analysis
Development Infrastructure	Docker CI/CD Cloudflare Terraform Linux WSL2
Agentic Engineering & AI Integration	Agentic workflows Claude Code Spec-Driven TDD Prompt Engineering
Workflow Automation	ERPNext Python tooling Retool → Expo migration
Languages	C/C++ Python JavaScript/TypeScript SQL Elixir
Embedded Systems	ESP32/ESP-IDF FreeRTOS NimBLE QP RTOS TinyFrame protocol
Web Development	ReactJS AstroJS Django REST API TailwindCSS Phoenix

AI Governance & Agentic Engineering

I treat AI as an agentic partner within a strict engineering framework, moving beyond "vibe-coding" to structured, audit-ready workflows:

- **Spec-Driven Development (SDD):** Prioritize context-heavy specifications to ensure alignment; AI-generated outputs are treated as hypotheses validated by human-led TDD.
- **Data Security:** Enforce mandatory sandboxing for all agentic projects; strictly prohibit credential access and utilize tool-based retrieval for scoped, anonymized data only.
- **Engineering Rigour:** All AI-assisted output undergoes rigorous human-in-the-loop review to ensure it meets enterprise quality and compliance benchmarks.
- **Privacy by Design:** AI workflows are architected to avoid direct database access; sensitive data remains behind scoped, tool-mediated retrieval with no standing credentials exposed to agents.

Job Target

Seeking a mid to senior software engineering role (remote preferred, hybrid max. 2 days/week on-site in Cape Town) with emphasis on embedded systems, backend services, or AI-assisted development workflows. Prefers small, focused teams with clear goals and engineering culture. Open to permanent or contract (3+ months) arrangements.